Name: _____

TOPIC TEST 1

Chapter 7: Loans and investments

- Time allowed: 45 minutes
- CAS calculators may be used.
- Part A: 5 multiple-choice questions (5 marks)
- Part B: 9 short-answer questions (27 marks)
- Part C: 3 analysis questions (18 marks)

Part A

Multiple-choice questions

5 multiple-choice questions
1 mark each: 5 marks
Circle the correct answer.

1 Sharon invests $1700 at 6.2% p.a. simple interest, added monthly. How much will she have in 9 months?

A $79.05

B $2648.60

C $1780.70

D $1779.05

E $80.70

[1 mark]

2 Sierra purchases an entertainment system worth $4700 on terms. She paid $900 deposit and 7.2% p.a. flat rate interest on the balance. She pays the balance in equal monthly instalments over 1 year. The amount of each monthly instalment is:

A $316.67

B $339.47

C $391.67

D $414.47

E $419.87

[1 mark]

© Cengage Learning Australia Pty Ltd 2014

3 The difference equation $t_{n+1} = t_n \times 1.075$ where $t_1 = 24\,000$, represents an investment of:

A \$24 000 at 7.5% p.a. compounded yearly

B \$75 000 at 24% p.a. compounded yearly

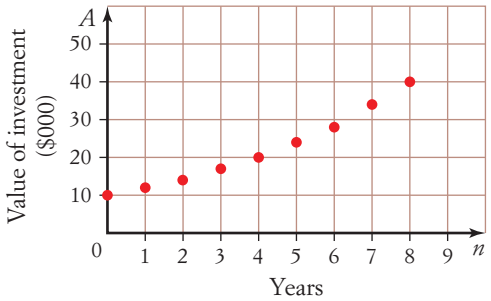
C \$24 000 at 1.075% p.a. compounded yearly

D \$24 000 at 10.75% p.a. compounded yearly

E \$1075 at 2.4% p.a. compounded yearly

[1 mark]

4 The following graph shows the increasing value of an investment over 8 years.



A rule for evaluating the value of the investment, in \$ after n years, could be:

A $10 \times (1.189)^n$

B $10\,000 \times (0.189)^n$

C $10 \times 0.189 \times n$

D $10\,000 \times (1.189)^n$

E $10\,000 \times 0.189 \times n$

[1 mark]

© Cengage Learning Australia Pty Ltd 2014

5 Neil and Jo each invest $64 000 at the following rates:

- Neil: 8.6% p.a. simple interest.
- Jo: 8.6% p.a. compounding quarterly

At the end of 4 years, Neil will have:

A $3006.40 more than Jo

B $3006.40 less than Jo

C the same amount of money as Jo

D $3932.50 more than Jo

E $3932.50 less than Jo

[1 mark]

Part B

Short-answer questions

9 short-answer questions
27 marks
Show your working where appropriate.

1 Find the interest paid for a week if Arthur borrows $18 000 at 9.2% flat rate of interest.

[2 marks]

2 Renee wants to earn $1000 interest in 2 years. She invests in an account earning 9.8% p.a. simple interest. How much would she need to invest? Answer to the nearest one hundred dollars.

[3 marks]

© Cengage Learning Australia Pty Ltd 2014

3 Michael purchases office furniture valued at $3700 on terms of $370 deposit and 52 fortnightly instalments of $71.56.

a How much interest does he pay in total?

b Calculate the annual flat rate of interest charged. Write your answer as a percentage correct to two decimal places.

[4 marks]

© Cengage Learning Australia Pty Ltd 2014

4 $2700 is invested at 8.9% p.a. compounded yearly.

a Using the formula $t_n = ar^{n-1}$, find the rule for the geometric sequence.

b What will be the value of the investment after 5 years?

[3 marks]

5 Bradley invests $8000 in an account paying 7.8% p.a. compounded quarterly.

a Write an expression involving the compound interest formula that can be used to find the value of his investment at the end of 3 years.

b Find the value of his investment at the end of 3 years.

[4 marks]

© Cengage Learning Australia Pty Ltd 2014

6 Narelle invests $87 000 in government bonds earning 6.25% p.a. compounding daily. How much interest will she earn in 90 days?

[3 marks]

7 How long will $45 000 need to be invested at 9.1% p.a. compounded 6-monthly in order to earn $10 000 interest?

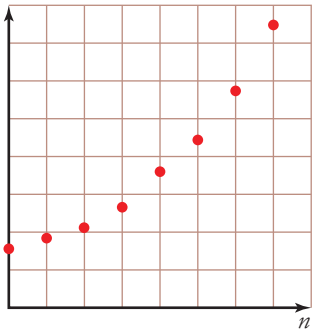
[3 marks]

© Cengage Learning Australia Pty Ltd 2014

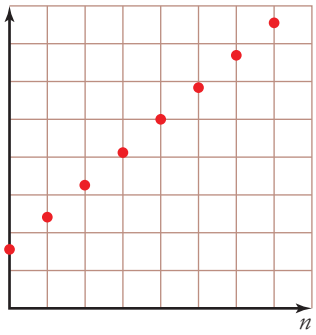
8 Patrice invests her money in an account earning 4.6% p.a. simple interest. She sketches a graph showing the value of her investment over 7 years.

Which of the following could represent her investment? Justify your answer.

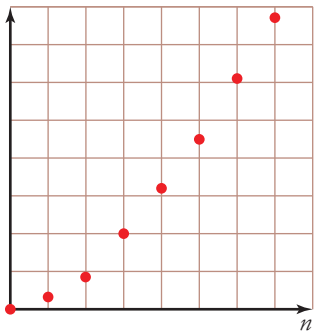
A



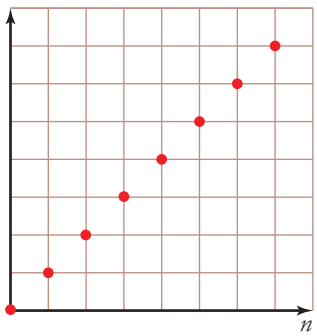
B



C



D



[3 marks]

9 Calculate the effective interest rate for an investment earning 8.2% compounded monthly. Answer as a percentage correct to one decimal place.

[2 marks]

© Cengage Learning Australia Pty Ltd 2014

Part C

Analysis questions

3 analysis questions
18 marks
Show your working where appropriate.

1 Melissa invests $16 000 at 6.8% p.a. compounding quarterly. In working out the interest earned in the third year, she writes the following expression where she calculates the interest earned in 3 years and subtracts it from the interest earned in 4 years.

$$16\,000 \times 1.068^4 - 16\,000 \times 1.068^3$$

a Explain why this expression is incorrect.

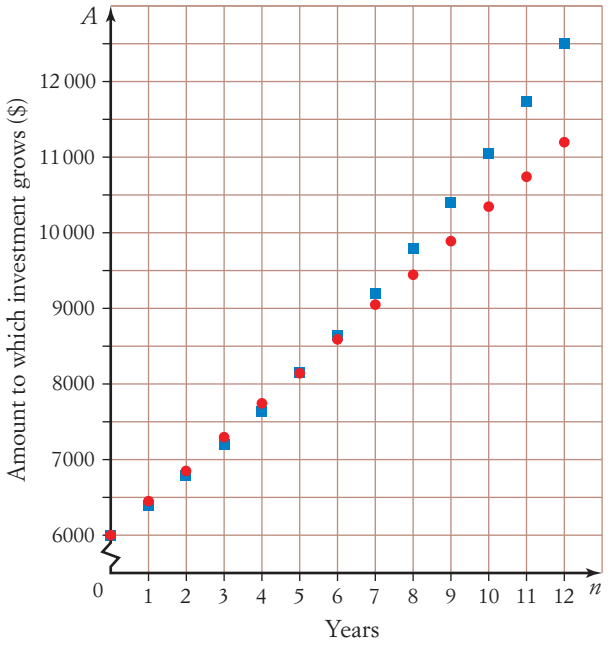
b Write the correct expression.

c Hence, calculate the interest earned in the third year.

[3 marks]

2 Easy Saver and Super Saver are two investment options at a bank. They both pay interest yearly, but Easy Saver pays simple interest and Super Saver pays compound interest.

The graph below shows the amount to which each investment has grown over a 12-year period when $6000 was initially invested in both.



a How can you tell that the red-dotted line represents Easy Saver?

© Cengage Learning Australia Pty Ltd 2014

b Which investment pays the highest annual interest rate, Easy Saver or Super Saver? Justify your answer.

c After 12 years, the Easy Saver investment has grown to $11 200.

i Find the interest paid yearly.

ii Find the yearly interest rate for the Easy Saver investment. Answer as a percentage correct to one decimal place.

d After 12 years, the Super Saver investment has grown to $12 500. Find the yearly interest rate for the Super Saver investment. Answer as a percentage correct to one decimal place.

e If someone was deciding between Easy Saver and Super Saver, what advice would you give them? Explain.

[10 marks]

© Cengage Learning Australia Pty Ltd 2014

3 Alex was looking to invest money over 7 years. He is comparing different investment accounts:

- Account 1: 8.6% p.a. simple interest.
- Account 2: 8.6% p.a. compounded annually.

a The accounts have the same interest rate per annum but Account 2 earns more interest. Explain why.

Alex was also considering another account:

- Account 3: 8.4% p.a. compounded monthly.

b Calculate the effective interest rate for Accounts 2 and 3.

c Which is the best option for Alex? Explain.

[5 marks]

[Total marks: 50]

© Cengage Learning Australia Pty Ltd 2014

Answers

Part A – Multiple-choice questions [1 mark each]

1 $I = ?, P = 1700, r = 6.2, n = \frac{9}{12}$

$$I = \frac{Prn}{100}$$
$$= \frac{1700 \times 6.2 \times \frac{9}{12}}{100}$$
$$= 79.05$$

$\therefore A = I + P$

$= 79.05 + 1700$

$= 1779.05$

$\therefore$ **D**

2 Balance $= 4700 - 900$

$= 3800$

$\therefore I = ?, P = 3800, r = 7.2, n = 1$

$$I = \frac{Prn}{100}$$
$$= \frac{3800 \times 7.2 \times 1}{100}$$
$$= 273.60$$

$\therefore$ Total to repay $= 273.60 + 3800$

$= 4073.60$

$\therefore$ Monthly instalment $= 4073.60 \div 12$

$= 339.466\ldots$

≈ 339.47

$\therefore$ **B**

3 $t_1 =$ initial value

$=$ the principal

$= 24\,000$

$$1.075 = 1 + \frac{\text{interest rate}}{100}$$
$$= 1 + \frac{7.5}{100}$$

$\therefore$ Interest rate $= 7.5\%$ p.a.

$\therefore$ **A**

© Cengage Learning Australia Pty Ltd 2014

4 Graph is curved, so we are using compound interest.

Therefore, it will take the form:

$A = P\left(1 + \frac{r}{100}\right)^n$ $\therefore$ not options C or E

Reading from the graph, the initial starting value (i.e. the principal) is \$10 000

$\therefore$ not option A.

Value in parentheses must be >1 $\therefore$ not option B.

$\therefore$ **D**

5 Neil:

$I = ?, P = 64\,000, r = 8.6, n = 4$

$$I = \frac{Prn}{100}$$
$$= \frac{64000 \times 8.6 \times 4}{100}$$
$$= 22\,016$$

$\therefore A = I + P$

$= 22\,016 + 64\,000$

$= 86\,016$

Jo:

$N = 4 \times 4 = 16$

$I(\%) = 8.6$

$PV = -64\,000$

$Pmt = 0$ $\therefore FV = 89\,948.496\ldots$

$FV = ?$ $\approx 89\,948.50$

$PpY = CpY = 4$

$PmtAt = END$

$\therefore$ Difference $= 89\,948.50 - 86\,016$

$= 3932.50$

$\therefore$ Neil will have \$3932.50 less than Jo.

$\therefore$ **E**

© Cengage Learning Australia Pty Ltd 2014

Part B – Short-answer questions

1 $I = ?, P = 18\,000, r = 9.2, n = \frac{1}{52}$ **[1 mark]**

$$I = \frac{Prn}{100}$$
$$= \frac{18\,000 \times 9.2 \times \frac{1}{52}}{100}$$
$$= 31.846\ldots$$
$$\approx 31.85$$ **[1 mark]**

$\therefore$ \$31.85 interest paid for a week.

2 $I = 1000, P = ?, r = 9.8, n = 2$ **[1 mark]**

$$I = \frac{Prn}{100}$$ **[1 mark]**
$$1000 = \frac{P \times 9.8 \times 2}{100}$$
$$\therefore P = 5102.040\ldots$$
$$\approx 5100$$

Therefore, Renee needs to invest \$5100. **[1 mark]**

3 a Total paid $= 74.56 \times 52 + 370$
$= 4091.12$ **[1 mark]**

$\therefore$ Interest $= 4091.12 - 3700$
$= 391.12$ **[1 mark]**

Therefore, he paid \$391.12 interest.

b Amount borrowed $= 3700 - 370$
$= 3330$ **[1 mark]**

$\therefore I = 391.12, P = 3330, r = ?, n = 2$

$$I = \frac{Prn}{100}$$
$$391.12 = \frac{3330 \times r \times 2}{100}$$
$$\therefore r = 5.872\ldots\%$$
$$= 5.87\%$$

Therefore, the flat rate of interest charged is 5.87% p.a. **[1 mark]**

© Cengage Learning Australia Pty Ltd 2014

4 a a = principal

$= 2700$

$r = 1 + \dfrac{\text{interest rate per time period}}{100}$

$= 1 + \dfrac{8.9}{100}$

$= 1.089$

$\therefore t_n = ar^{n-1}$

$= 2700 \times 1.089^{n-1}$ **[1 mark]**

b After 5 years, $n = 6$ **[1 mark]**

$\therefore t_6 = 2700 \times 1.089^{6-1}$

$= 4135.263\ldots$

≈ 4135.26

$\therefore$ The value of the investment after 5 years will be \$4135.26. **[1 mark]**

5 a $A = ?, P = 8000, r = \dfrac{7.8}{4} = 1.95\%$ per quarter, $n = 3 \times 4 = 12$ quarters

[1 mark for *r* and 1 mark for *n*]

$A = P\left(1 + \dfrac{r}{100}\right)^n$

$= 8000\left(1 + \dfrac{1.95}{100}\right)^{12}$ **[1 mark]**

Should also accept 8000×1.0195^{12}

b $A = 10\,086.413\ldots$

$\approx \$10\,086.41$

His investment will be worth \$10 086.41 at the end of 3 years. **[1 mark]**

6 $N = 90$ **[1 mark for writing known values either in list or in formula.]**

$I(\%) = 6.25$

$PV = -87\,000$

$Pmt = 0$ $\quad \therefore FV = 88\,351.021\ldots$

$FV = ?$ $\quad \approx 88\,351.02$ **[1 mark for *FV*]**

$PpY = \left.\begin{matrix} \\ \\ \end{matrix}\right\} 365$ $\quad \therefore I = 88\,351.02 - 87\,000$

$CpY = \left.\begin{matrix} \\ \\ \end{matrix}\right\}$

$PmtAt = END$ $\quad = 1351.02$

Therefore, she will earn \$1351.02 interest in 90 days. **[1 mark]**

© Cengage Learning Australia Pty Ltd 2014

7 $N = ?$ **[1 mark for writing known values either in list or in formula.]**

$I(\%) = 9.1$

$PV = -45\,000$

$Pmt = 0$ $\therefore N = 4.50\ldots$ 6 months **[1 mark for *N*]**

$FV = 55\,000$

$PpY = \left.\begin{matrix}\\ \\ \end{matrix}\right\} 2$

$CpY = \left.\begin{matrix}\\ \\ \end{matrix}\right\}$ $\therefore$ take 5 lots of 6 months

$PmtAt = END$

Therefore, it will take $2\frac{1}{2}$ years. **[1 mark]**

8 Graph B **[1 mark]**

It is simple interest, so it must be a straight-line graph $\therefore$ B or D.

[1 mark for this reason]

We want the value of her investment, so, at $n = 0$, the value is the principal. Therefore, the graph must start above the zero on the y-axis $\therefore$ B.

[1 mark for this reason]

9 $I = 8.2\% = 0.082,\ n = 12$

$e = (1 + \frac{i}{n})^n - 1$

$= (1 + \frac{0.082}{12})^{12} - 1$ **[1 mark]**

$= 0.0851\ldots$

$\approx 8.5\%$ **[1 mark]**

Therefore, the effective interest rate is 8.5%.

Part C – Analysis questions

1 a It has the interest rate and years as compounding yearly, but it should be compounding quarterly. **[1 mark]**

b $16\,000 \times 1.017^{16} - 16\,000 \times 1.017^{12}$ **[1 mark]**

c $1366.277\ldots \approx 1366.28$

Therefore, the interest earned in the third year is $1366.28. **[1 mark]**

2 a It is a straight line which represents a simple interest investment. **[1 mark]**

b Easy Saver **[1 mark]**

Initially, Easy Saver earns the most interest. This can be seen by the fact that the red-dotted line is initially above the line marked with squares. **[1 mark]**

© Cengage Learning Australia Pty Ltd 2014

c i Total interest $= 11\,200 - 6000$

$= 5200$ **[1 mark]**

$\therefore$ Interest per year $= 5200 \div 12$

$= 433.333\ldots$

≈ 433.33

$\therefore$ Interest paid per year is \$433.33. **[1 mark]**

ii $I = 433.33, P = 6000, r = ?, n = 1$ **[1 mark for using correct *I* and *P*]**

$$I = \frac{Prn}{100}$$

$$433.33 = \frac{6000 \times r \times 1}{100}$$

$\therefore r = 7.22\ldots\%$ p.a.

$\approx 7.2\%$ p.a. **[1 mark]**

d $N = 12$ **[1 mark for writing known values either in list or in formula.]**

$I(\%) = ?$

$PV = -6000$

$Pmt = 0$ $\therefore r = 6.30\ldots$

$FV = 12\,500$ $\approx 6.3\%$ p.a.

$PpY = \left.\begin{matrix} \\ \\ \end{matrix}\right\}$
$CpY = \left.\begin{matrix} \\ \\ \end{matrix}\right\} 1$

$PmtAt = END$

Therefore, the interest rate for the Super Saver investment is 6.3% p.a. **[1 mark]**

e If investing for more than 5 years, the Super Saver account is the best option. Otherwise, Easy Saver is the best option. This can be seen on the graph as below 5 years the Easy Saver graph is above the Super Saver but from 5 years it is reversed. **[1 mark]**

3 a The interest earned in Account 2 is compounded, which means the interest earned each year gets added to the principal. Thus, interest is earned on the previous year's interest as well as on the principal. In contrast, Account 1 earns simple interest, which means interest is earned on the principal only. **[1 mark]**

b Account 2: $e = \left(1 + \frac{0.086}{1}\right)^1 - 1 = 0.086$

$= 8.6\%$ **[1 mark]**

Account 3: $e = \left(1 + \frac{0.084}{12}\right)^{12} - 1 = 0.08731\ldots$

$\approx 8.73\%$ **[1 mark]**

c Account 3 is the best option. **[1 mark]**

It has the higher effective interest rate and will therefore pay higher interest over time. **[1 mark]**

© Cengage Learning Australia Pty Ltd 2014